Same Text, Different Filename: Are LLM Judges Biased?

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Abstract

Yes. In 2026, leading economic journals started using Refine.ink in the technical quality assessment of submissions before publication. I study whether renaming a file changes the perception of the document's quality by LLM judges. I find that three frontier LLMs score the same document differently if you change the document's filename. A person-related label scores higher than no label; there is no consistent preference of own- or rival-based label, nor a penalty for an Asian name, but there is a context-specific bias in assessments whose direction and size vary with both document and label. With LLM-written documents, some self-premium can be detected, but a significant context-dependent label bias remains. Whether to combat it is a question of the costs of evaluation errors.

Keywords: Large Language Models; LLM-as-a-Judge; Peer Review; Algorithmic Bias

JEL Classification: C91, J71

*Cardiff University, popovs@cardiff.ac.uk. The usual disclaimer applies. Studies were preregistered at (1) https://aspredicted.org/sq9mp2.pdf, then updated at https://aspredicted.org/qj59r2.pdf and https://aspredicted.org/vp8ez6.pdf and (2) https://aspredicted.org/82v8c4.pdf. LLMs helped me with coding and preparing the pre-registrations.

Introduction

In 2026, Large Language Models (LLMs) are used by scientists in developing both the content and the text of their studies. Since 2026 EU law (Article 50, European Parliament and Council of the European Union, 2024) requires LLMs to mark synthetic output in machine-readable form. Verification of these marks should allow editors to detect LLM use, and if the journal does not allow it, prosecute the violators (indeed, arxiv.org has a one-strike policy against unchecked LLM use, Stan, 2026).

On the other hand, direct use of LLMs in processing structured text that made it omnipresent in coding environments allowed easy proofreading of academic papers, and one of the refereeing startups, Refine.ink, was officially adopted by editorial boards of leading Economics journals as an intermediary step of evaluation of manuscripts before publication (Golub, 2026). The combination of facilitated detectability of LLM use, use of LLMs for judgment calls about textual quality, and competition between LLM models creates a sensitive point: would an LLM underlying an automated assessing tool be able to assess the document ignoring whether it was treated by the tool or underlying LLM?

This bias does not have to be malicious: treatment of manuscripts by LLMs means maximizing a value function subject to constraints, and different LLMs differ in values and constraints, so an LLM that used one approach will deem something that maximized a different approach suboptimal from their own perspective; whereas a document that was already treated by the assessing LLM will look close to optimal just because it was assessed by that LLM (Deutsch et al., 2022). There is literature about LLM bias; this paper differs by asking whether the difference in the *filename* would introduce a difference in assessment.

To find out, I take 4 documents that I wrote myself, convert to textual Markdown format to minimize metadata embedded in more sophisticated formats, and manipulate the filename before asking LLMs to assess the quality of the document's writing. This is not the same target as the target of verification tools such as Refine.ink, but writing quality assessment is LLMs' core competence, and therefore any deviations we detect would be informative about the dangers for more specialized tasks.

I find a small label bias, but more importantly, I find a significant document-specific bias: it's comparable to the persistent bias. This motivates me to run another study on LLM-generated texts to see whether this document-specific bias is an artifact of a small number of human-written texts in my corpus. With 90 LLM-generated texts, I find the same label bias in the same assessors. This context-specific bias has not been studied before, and its impact is at least as big as the permanent bias. This has implications for fighting bias in non-LLM assessment: if humans exhibit context-specific bias in assessment, this bias might not only conceal the persistent bias by making data noisier, but also would not be affected by persistent-bias lowering techniques.

The literature review is followed by the experimental description. Then studies' results are presented, which are then followed by the discussion and conclusion.

Existing Literature

The bias of anonymous assessment of living referees in Economics is well documented. Blank (1991) argued for double-blind review in times before working papers were posted online. Huber et al. (2022) ran an experiment on referees; apparently, Nobel prize winner signing a paper immediately improves its novelty, relevance, and clarity of writing, even if you don't change the text. Formatting affects perception: James and Kjorstad (2026) find that LaTeX-formatted papers are perceived as of better quality. To test for biases, experiments with label manipulations are numerous: Bertrand and Mullainathan (2004) manipulated names on CVs and found that HRs respond to some names more than to others (Bertrand and Duflo, 2017, provides a survey). This paper uses the same approach to manipulate the filename without changing the text to see if LLM's assessment of the quality of writing is different. Replacing biased humans with unbiased computers is tantalising (Kim et al., 2026, King and Worthington, 2026, Mizumoto and Eguchi, 2023, Pataranutaporn et al., 2026, Russo Latona et al., 2024, Thakkar et al., 2026). LLMs however are not unbiased.

The sensitivity of LLMs to manipulations of texts and prompts is well researched (Lior et al., 2026, Robinson and Burden, 2025). Bias towards human authors is detected by above studies and others (Haverals and Martin, 2025, Sun et al., 2026). In that literature, Saraf et al. (2025) run an experiment similar to mine except their texts are LLM-generated, and the label is part of their call to the LLM judge; Germani and Spitale (2025) do the same thing, but also find bias against Chinese-labelled sources, and one of my hypotheses tests that bias (see also Nghiem et al., 2024). Abolghasemi et al. (2025) uses texts produced by humans and LLMs, and submits authorship labels as part of the text tag. In my first experiment, I was the author of all texts, so the difference between different texts is not from which LLM it came, but only from any bias that LLMs develop from seeing the document's filename; I detect some self-preference in my second study. Rospocher (2026) shows that author identity (bibliometrics, institution tier) affects LLM judge decisions.

Papers above, as others (Chae et al., 2026, Deutsch et al., 2022, Wataoka et al., 2024) do, find LLM bias towards their own writing, but I show that some bias, albeit less pronounced and with ambiguous direction, remains even if the true writing is not of their own: it's not just about how things are written, it's about the label. Panickssery et al. (2024) confirm that LLMs could recognize their own writing even before the watermark was introduced as a response to EU regulation. My label exploration study is different from these because it submits human-written text (written by me), ruling out the Deutsch et al. (2022) explanation that LLMs demonstrate self-bias by design.

Experiment

Pipeline I use 4 English language texts that were not available online prior to this experiment that I wrote at different times: one overview of the junior Welsh chess scene (Chess at 1167 words), one assessment for a teaching certificate (Assignment at 2915 words), one sample answer to an exam question (Cournot at 340 words), and one segment from an unwritten public-audience book (Book at 1156 words). I saved their MD5 and SHA256 hashes after removing identifying information. These documents are available for replication by request. They are saved as `.md` files to minimise the influence of file tags and other excessive information of more sophisticated formats.

I use 3 widely used LLMs to evaluate these texts for the quality of writing. I took these texts in particular because they are not very good; I need to avoid the ceiling effect. I ask each LLM

```
Evaluate the attached document for quality of writing on a scale from 0 to 10. Use one decimal place. Do not elaborate, just return the number.
```

Each LLM separately assessed 6 versions of the file with identical content but different filenames:

`Document.md`	`Document_from_Sergey.md`
`Document_from_Chen.md`	`Document_from_ChatGPT.md`
`Document_from_Claude.md`	`Document_from_Gemini.md`

I use OpenRouter to access LLMs, which proved cheaper than buying subscriptions. Anthropic Claude designed my R code for API calls.

Hypotheses As registered at AsPredicted, I test 5 hypotheses:

H1 Documents labelled as written by an LLM (ChatGPT, Claude, or Gemini) receive lower scores, on average, than the same documents with a neutral or Sergey anchor.

H2 For each evaluating model, a document labelled as written by that model's own family receives a higher score than the same document labelled as written by either rival model.

H3 For the same evaluating models, documents labelled as written by a rival model receive lower scores than the same documents with a neutral filename.

H4 and H4b Documents labelled with an Asian-coded human name receive lower scores on average than the same documents labelled with a non-Asian human or neutral filename. As a secondary comparison, I also report an Asian-coded label (Chen) against non-Asian (Sergey).

To test these hypotheses, I run an OLS regression of scores reported by models on dummy variables of the used document and the used label (neutral being the reference category), for each judge model separately. Two-sided tests help me evaluate my hypotheses.

Pilot I used a free Ling 3.0 Flash VL `inclusionai/ling-3.0-flash-vl:free` model via OpenRouter which allowed 484 calls before my daily cap triggered (so the last call did not return a result). Since the model is from neither of my 3 providers, this allows me to estimate power needed for my human-vs-LLM evaluation. The preliminary estimates showed H1's effect of -0.171 with 0.050 standard error and p-value of 0.0006; the model punishes LLM label against human label by 0.239 (SE 0.066, p-value 0.0003). These effects are small compared to Saraf et al. (2025): they estimate shifts of ≈ 12 percentage points, if converted to a comparable 0–10 scale. The residual standard deviation of 0.536 suggests that if I want to detect a 0.1 size effect with a power of 90%, I need 63 repetitions to test H1, 114 repetitions[1] for H2-H4, and 152 for H4b (its k is 2). I therefore am running 152 requests per judge per document per label.

Human-written Texts

After the ethics approval of Cardiff Business School's SREC panel was obtained (#3355), on October 27, 2026, I was able to run the code on `anthropic/claude-sonnet-5.5`, `openai/gpt-6.1-sol`, and `google/gemini-4-argon`, three models of similar capability from different providers 152 repetitions per document-label yields 3648 calls per model, or 10944 observations. Order of documents and labels is randomized in a saved schedule. Every call is an independent conversation, temperature is default and since no reasoning is recorded or requested, max tokens is capped at 1000 (never reached). Routing was managed by OpenRouter; Gemini 4 Argon got split between two Google routes, with no statistically significant differences in outcomes. The cost was about USD 36. The first run, executed in parallel, led to 5 problems in the data due to attempts to write in the same file simultaneously, so I reran the failed 5 trials. The code and outcomes can be obtained from the repository: `https://github.com/sergeyvpopov/TextEval`.

Unfortunately, OpenRouter cuts off the filenames of uploaded files, which made me submit the filename as part of the text. Another approach would be to upload a .pdf and send a link for processing by the LLM, but .pdfs have more metadata, so I decided to use the filename in text instead. All 10944 replies were valid, realised residual SDs were far below the pilot's, so I include the minimum detectable effect at 80% power (which I aimed to be 0.1 in power calculations).

All 10,944 marks lie between 2.5 and 6.8, so the models use less than half of the 0–10 scale, and each uses a narrower band: GPT-6.1 Sol uses [4.6,6.5], Claude Sonnet 5.5 uses [2.5,6.5] and Gemini 4 Argon uses [4.2,6.8]. Against this, a typical label effect of 0.1–0.2 points is 1–2% of the scale, but it is not small relative to the spread of the marks. Within one document and label, 95% of single calls lie within $[-0.32, 0.28]$ of the cell mean for Sol, $[-0.52, +0.72]$ for Sonnet and $[-0.53, +0.50]$ for

[1]The formula I used is $N = \frac{k\sigma^2(z_{1-\alpha/2}+z_{1-\beta})^2}{\Delta^2}$, where $\Delta = 0.1$, $\sigma = 0.536$, $\alpha = 0.05$, power $1-\beta = 0.9$, and $k = 3\left(\frac{1}{3}\right)^2 + 2\left(\frac{1}{2}\right)^2$ for testing H1 vs $k = 1 + 2\left(\frac{1}{2}\right)^2$ to test H2-H4 (this depends on the weights of the coefficients in the test); divide by 4 to obtain the number of repetitions.

Argon. The largest label differences within a single document reach 0.39 points for Sol (Book), 0.85 for Sonnet (Book) and 0.68 for Argon (Book), which for Sonnet is about 44% of the gap between its highest and lowest document means (1.9 points). Marks are also coarse: Sonnet's three most frequent values (3.5, 4.5 and 5.8) account for 76% of its replies, and an average bias represents a change in probabilities, not a shift in values.

Table 1: Registered contrasts by model; fixed effects for documents. MDE_{80} is the minimum detectable effect at 80% power.

Hypothesis	Model	Estimate	SE	95% CI	MDE_{80}	p (Holm)
H1	openai	-0.092	0.006	[-0.104, -0.081]	0.019	<0.001
H1	anthropic	0.123	0.012	[0.099, 0.147]	0.039	<0.001
H1	google	-0.203	0.010	[-0.224, -0.183]	0.034	<0.001
H2	openai	-0.037	0.009	[-0.054, -0.020]	0.028	<0.001
H2	anthropic	-0.099	0.020	[-0.137, -0.060]	0.064	<0.001
H2	google	0.046	0.014	[0.019, 0.073]	0.045	<0.001
H3	openai	-0.021	0.008	[-0.037, -0.006]	0.026	0.007
H3	anthropic	0.279	0.015	[0.249, 0.309]	0.050	<0.001
H3	google	-0.175	0.014	[-0.203, -0.147]	0.046	<0.001
H4	openai	0.058	0.006	[0.045, 0.070]	0.021	<0.001
H4	anthropic	0.073	0.015	[0.043, 0.103]	0.049	<0.001
H4	google	-0.010	0.014	[-0.037, 0.018]	0.046	0.492
H4b	openai	-0.001	0.007	[-0.015, 0.014]	0.024	0.911
H4b	anthropic	-0.050	0.018	[-0.085, -0.016]	0.057	0.008
H4b	google	-0.053	0.016	[-0.085, -0.022]	0.052	0.003

Estimation is a model-specific OLS regression as preregistered, with document fixed effects and HC3 standard errors (can't cluster because very few documents). I used Holm correction for multiple testing, with $\alpha = 0.05$. The hypotheses are linear tests of dummy variable coefficients.

H1 (LLM-label penalty) LLM penalty is visible but not uniform; Anthropic on average prefers LLM-labelled output to human-labelled output. The findings are consistent with expected; the impact is not large, 0.1–0.2 on a 10-point scale on average. This is smaller than what Saraf et al. (2025) finds.

H2 (own-family premium) There is no consistent own-family effect. Argon shows self-preference (+0.046), others show small self-deprecation. It could be driven by the fact that my documents are not really written by LLMs; I did not test the bias on LLM-generated documents.

H3 (rival penalty) The OpenAI model punishes other models; the Anthropic model praises them, beyond what could be found due to self-deprecation; the Google model punishes them; the effects are not large. This could be driven by the fact that the documents are human-generated.

H4 (Chen vs Sergey+Neutral) and H4b (Chen vs Sergey) Anti-Asian bias of Germani and Spitale (2025) is borderline visible; compared to Sergey label, Chen obtains about 0.05 lower score.

Table 2: Exploratory analysis by document; differences in documents affect the biases studied by the hypotheses. Star denotes a 1% significant effect, unadjusted for multiple testing; brackets contain HC3 standard errors.

Model	Document	H1	H2	H3	H4	H4b
GPT-6.1 Sol	Assignment	-0.07*	0.03	-0.03*	0.11*	0.07*
		(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
	Book	-0.23*	-0.14*	-0.12*	-0.02	-0.08*
		(0.01)	(0.02)	(0.02)	(0.02)	(0.02)
	Chess	-0.05*	-0.02	0.01	0.04*	-0.01
		(0.01)	(0.01)	(0.01)	(0.01)	(0.01)
	Cournot	-0.02	-0.01	0.05*	0.10*	0.03
		(0.01)	(0.02)	(0.02)	(0.01)	(0.02)
Claude Sonnet 5.5	Assignment	-0.15*	-0.09*	-0.11*	-0.20*	-0.21*
		(0.01)	(0.01)	(0.02)	(0.01)	(0.02)
	Book	0.23*	-0.56*	0.52*	0.17*	0.07
		(0.02)	(0.02)	(0.02)	(0.04)	(0.05)
	Chess	0.19*	-0.49*	0.77*	0.14*	-0.28*
		(0.01)	(0.03)	(0.02)	(0.02)	(0.02)
	Cournot	0.22*	0.75*	-0.07*	0.19*	0.22*
		(0.02)	(0.03)	(0.02)	(0.03)	(0.03)
Gemini 4 Argon	Assignment	-0.20*	0.07*	-0.17*	-0.06*	-0.11*
		(0.02)	(0.03)	(0.03)	(0.02)	(0.02)
	Book	-0.35*	0.04	-0.24*	-0.23*	-0.35*
		(0.02)	(0.03)	(0.03)	(0.03)	(0.03)
	Chess	-0.07*	0.03	-0.01	0.08*	0.01
		(0.01)	(0.01)	(0.02)	(0.02)	(0.02)
	Cournot	-0.20*	0.05	-0.28*	0.17*	0.24*
		(0.03)	(0.04)	(0.04)	(0.03)	(0.04)

There are probably other biases (gender, specific names, etc) at play, and future research can explore these venues. As an exploratory analysis, I also run a sign test: 12 out of 13 significant tests remain significant with the same sign; 0 out of 2 insignificant tests become significant.

Documents are of different lengths (Assignment is longest, Cournot is shortest), written at different times, and aimed at different audiences. This makes marks for different texts different; but, as Table 2 shows, biases are not document-neutral. Own-family effects seem to be unstable: depending on the document, Sonnet can exhibit both self-deprecation (-0.56 for Book and -0.49 for Chess) and self-aggrandizing (Cournot +0.75); Gemini's effect is borderline zero. Note that for the H4 anti-Asian bias the zero overall bias of Table 1 averages large biases for individual documents.

Like any study, this study has limitations. I have only 4 documents; their advantage is that I know they were done without any LLM assistance, and LLMs were not trained on them before the experiment. I test 6 labels; they were chosen ad hoc, and a different label could exhibit a signifi-

cant difference. The filename had to be sent via OpenRouter as a part of the prompt; the effect of uploading a file might differ. Exploratory analysis (one model, one document) indicates that uploading matters. Indeed, the Claude run with Cournot via OpenRouter yielded a mark of 3.56 for no label with an SD of 0.22, and a mark of 4.25 for a Claude label with an SD of 0.42. A similar run via Claude (Opus opening Sonnet instances) yielded different marks when the text was provided as with OpenRouter due to differences in harness, tools and context: 5.57 SD 0.23 (neutral label) and 5.58 SD 0.28 (Claude label). The same run with uploaded files changed marks to 4.77 SD 0.43 (neutral) and 4.49 SD 0.11 (Claude). The difference in label bias can be attributed to reading via a tool instead of submitting as part of the prompt, so biases are not neutral to the delivery method.

The LLMs I studied score differently identical documents when only the implied filename changes. A file with a human-name label scores higher than an unlabelled document. LLM-labelled models score lower than no label for Sol, higher for Sonnet, and lower for Argon. I find no consistent preference for oneself for any of three models; Asian-coded name label does not seem to carry a large penalty compared to Sergey.

Nevertheless, biases remain. The biases seem most pronounced when interacted with documents. For instance, Gemini finds little difference between Chen and Sergey labels on average, but that decomposes into a +0.24 bonus for Chen for the Cournot document averaged with -0.35 penalty when the Book document is analyzed. Biases are not document-neutral, context and delivery of manuscripts matter too; insignificance of the bias on average does not mean the bias is not there on the document level. Whether to tolerate the bias is an economic question, which requires knowledge of the costs of decision errors of both kinds, for both decisionmakers and applicants.

Bias Decomposition

To see how much bias is persistent, and how much is context-sensitive, I pre-registered a second study on AsPredicted #321045 on October 29, 2026, to generate 10 briefs in 3 genres and feed them to 3 LLMs to generate texts for analysis. This obtains 90 texts, which I then label as above, plus two labels, `_v2` and `_from_Catherine`. The first study informed that LLMs are relatively persistent in their estimates, so there is little statistical value in running more than 12 evaluations per text. I ask:

Q1 Does a label effect exist? My previous study found it for human writing.

Q2 How large is it? Study 1 found a persistent bias as large as Anthropic preference for rival labels of $+0.279$, and a persistent penalty in Google evaluation of rival labels of -0.175. Document-specific bias was even larger: Claude gave a 0.77 higher evaluation to texts labeled by rivals when working with the Chess text, and Gemini penalised the Cournot text written by rivals by -0.28. This study adds another human name and a version label.

Q3 What is the proportion of persistent bias across documents? The exploratory analysis of the previous study calculates the label bias conditional on the document, and then calculates the average bias across labels for each LLM judge. This yields a persistent bias SD across documents of 0.07 for Sol, 0.11 for Sonnet, and 0.11 for Gemini; the document-specific part is 0.06 for Sol, 0.22 for Sonnet, and 0.09 for Argon. This means persistent bias is 57% of realised bias for Sol, 19% for Sonnet, and 62% for Argon. These were not very reliable based on 4 texts; this study generates more texts, but uses LLM generation for this, and asks how much these shares change.

The preregistered hypotheses (all tests two-sided) are

H5 self-preference: own family scored higher under neutral label, predicted positive;

H6 own label on own text bias: predicted positive;

H7 LLM penalty: LLM labels score lower than human name labels, predicted negative.

Table 3: Named persistent biases, Study 2 (H5–H7), by judge. Unit of analysis is the brief (30 briefs, 29 degrees of freedom); p values are Holm-adjusted across the three judges within each hypothesis. H5: score a judge gives its own familys text minus the score predicted from judge and generator effects fitted on the other six judgegenerator pairs, under the neutral label; H6: own-family label minus the mean of the two rival LLM labels, on own-family text; H7: LLM labels minus human names. Briefs shows positive/negative contrasts. MDE_{80} is the minimum detectable effect at 80% power.

Hypothesis	Judge	Estimate	SE	95% CI	Briefs	MDE_{80}	p (Holm)
H5	GPT-6.1 Sol	0.338	0.049	[0.238, 0.437]	29/1	0.165	<0.001
	Claude Sonnet 5.5	-0.281	0.068	[-0.419, -0.142]	5/25	0.230	<0.001
	Gemini 4 Argon	0.551	0.083	[0.382, 0.721]	30/0	0.281	<0.001
H6	GPT-6.1 Sol	0.018	0.006	[0.005, 0.032]	20/10	0.022	0.016
	Claude Sonnet 5.5	0.019	0.024	[-0.030, 0.067]	13/16	0.081	0.436
	Gemini 4 Argon	0.157	0.022	[0.112, 0.203]	28/2	0.076	<0.001
H7	GPT-6.1 Sol	-0.111	0.007	[-0.125, -0.098]	0/30	0.023	<0.001
	Claude Sonnet 5.5	-0.120	0.018	[-0.157, -0.082]	4/26	0.062	<0.001
	Gemini 4 Argon	-0.127	0.010	[-0.147, -0.108]	0/30	0.033	<0.001

The run cost USD 47.66, returned 25,920 valid responses, with 12 repetitions for every judge, and took 20 hours, much of which consisted of delays between calls to not violate ToS of OpenRouter. The remaining technical details were similar to the first study.

Self-preference is confirmed (acknowledging Sonnet's self-deprecation); human preference is confirmed. To see the variation across documents analogously to Table 2, observe Figure 1. Three columns represent valuations by three judges. Each row shows evaluations with each label (0 for no

label, 1 for `_v2`, 2–4 for OpenAI, Anthropic, and Google, and 5–7 for Sergey, Chen, and Catherine). Documents are coded by author (O for OpenAI, A for Anthropic and G for Google) followed by a Cyrillic brief letter (see Appendix A for details of brief topics). Label variation in a row shows the label effect; label variation across documents shows variation across briefs and authors. The difference in briefs is seen as difference in writing quality. The brief topics are benign, with zero references to say crime to minimize the risk of getting my account banned; a sensitive topic might be interpreted by an LLM judge as a low (or high!) quality document. I don't study this, but it is a clear venue for future researchers, and has clear implications for LLM-based evaluations of the quality of mathematical analysis in applied theory papers.

To conclude, in Table 4 I report the share of the persistent bias variance in total bias variance. LLM-written texts exhibit similar patterns to human-written texts; it is not something specific to human writing that drives context-specific biases, it's the interaction with labels.

Table 4: Persistent share of label-related variance (percent). Corrected shares adjust for finite-repetition noise; 95% cluster-bootstrap percentile intervals in brackets (2,000 resamples). Q3 columns use between-unit products (documents for Study 1, briefs for Study 2) as preregistered.

Judge	Human texts (corrected)	LLM-written (corrected)	Human texts (Q3)	LLM-written (Q3)
GPT-6.1 Sol	57.3% [50.8, 94.9]	63.6% [58.9, 70.4]	43.0% [34.4, 93.2]	63.1% [58.4, 69.7]
Claude Sonnet 5.5	19.3% [9.5, 87.7]	17.6% [12.9, 26.2]	-7.5% [-20.7, 83.6]	16.6% [12.1, 25.3]
Gemini 4 Argon	62.4% [53.1, 92.1]	62.7% [57.8, 69.1]	49.8% [37.5, 89.5]	62.3% [57.1, 69.0]

Conclusion

I find that filename labels introduce bias. The persistent bias is small but detectable: LLMs prefer people over LLMs as authors, and absence of informative label is often worse than both. However, besides the persistent bias, there is context-specific bias, whose impact is frequently stronger than the persistent part, and whose variability masks the significance of the bias. I detect it in LLM judges; it could be in human judges too.

The biases I document might decrease with time as models become more sophisticated. To combat them, one could reduce the label effect to no-label bias by scrambling any metadata before the analysis. The invisible watermark, LLM verbiage ("load-bearing", "spine", em-dashes), and cadence are harder to get rid of, and they could be interacting with documents too. Some of the bias can be averaged out by using multiple LLMs multiple times and averaging the outcomes; such an approach is unlikely to work with human judges. Future research can help quantify the machinery behind the context-specific bias, which hopefully will inform ways of combating it.

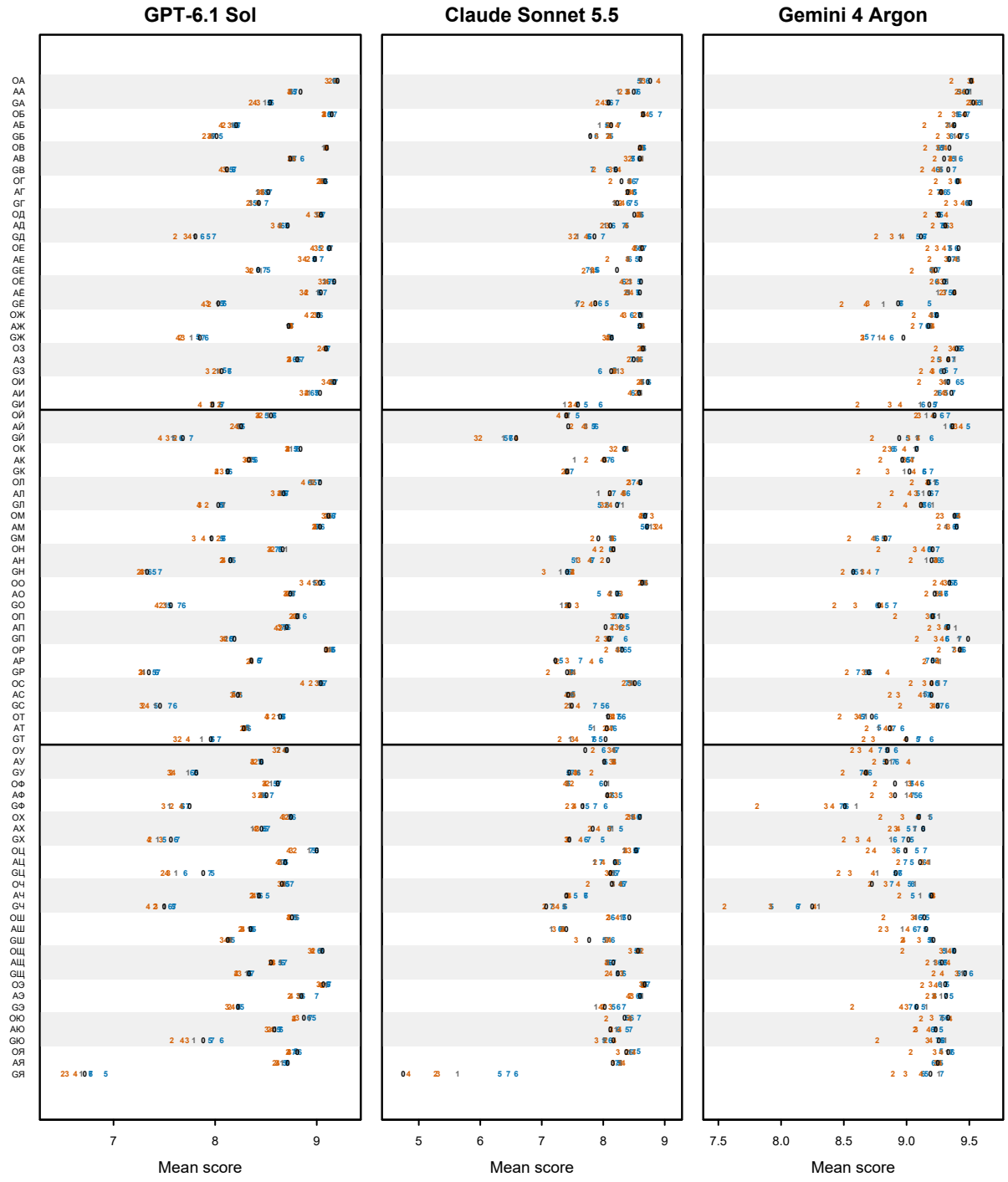


Figure 1: Scores for each of the 90 texts under each label, by judge (Study 2). Rows are texts: the first third are academic texts, the second general-audience essays, the last reports; lines separate. Each is coded by writer (O = OpenAI, A = Anthropic, G = Gemini) followed by a Cyrillic brief letter (key in the Table 5). Each digit is one label, placed at the text's mean score (over the 12 repetitions) under that label: 0 = neutral label (black); 1 = _v2 (grey); 2, 3, 4 = ChatGPT, Claude, Gemini labels (orange); 5, 6, 7 = Sergey, Chen, Catherine (blue). Each panel spans that judge's lowest to highest text-by-label mean.

Table 5: Key to the brief codes in Figure 1. The confusing enumeration of briefs was my mistake when generating manuscripts; I decided against manual renaming.

Code	Brief	Description
а	B01CournotVsStackelberg	academic text: Cournot vs Stackelberg
б	B02Eignevalues	academic text on eigenvalues
в	B02InvisibleHand	academic text on Invisible Hand
г	B04Money	academic text on money neutrality
д	B05BehaviouralFinance	academic text on behavioural finance
е	B06WritingAPaper	academic text on paper writing
ё	B07WritingaSOP	academic text: PhD purpose statement
ж	B12Biases	academic text on LLM bias
з	B18Syllabus	academic text: machine-learning course outline
и	B22InflationUnemployment	academic text: inflation and unemployment
й	B08Appeal	general essay: parking appeal letter
к	B09Chess	general essay on UK chess
л	B10Mars	general essay on Mars colony
м	B11LLMUse	general essay on LLM self-education
н	B13ChessAd	general essay: chess for children
о	B15SourcesOfCreativity	general essay on creativity sources
п	B21Brexit	general essay on Brexit
р	B25Lovecraft	general essay on Lovecraft's pantheon
с	B26Coffee	general essay: espresso versus jezve
т	B30Fields	general essay: employable European fields
у	B16RealEstate	report on Welsh real estate
ф	B17Travel	report: Spanish scenic road trip
х	B17Universities	report on German higher education
ц	B19Sleep	report on sleep deprivation
ч	B20CookingPlan	report: week of student meals
ш	B23Carcassonne	report: Carcassonne and similar games
щ	B24Bardic	report: Welsh bards versus troubadours
э	B27Painting	report on Galileo's scientific role
ю	B28Art	report on Basquiat's artistic role
я	B29Tales	report: Welsh tale tropes

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